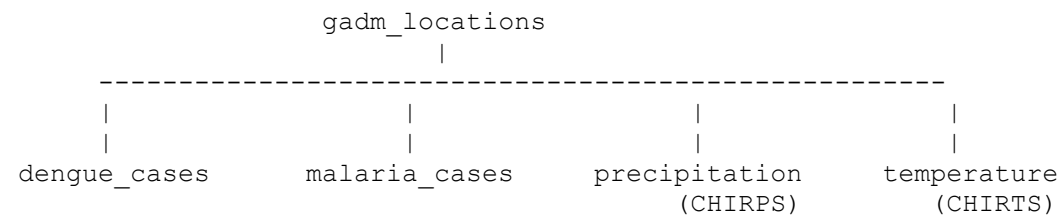


Database Name

brazil_disease_analysis

Database Tables



The **gadm_locations** table is the central table because every disease case, precipitation record, and temperature record belongs to a geographical location.

Table 1: gadm_locations

This table stores the geographical information from the GADM dataset.

Column	Data Type	Description
location_id	INT AUTO_INCREMENT PRIMARY KEY	Unique location ID
country	VARCHAR(100)	Country name
state_code	VARCHAR(10)	State abbreviation
state_name	VARCHAR(100)	State name
municipality	VARCHAR(150)	Municipality name
geometry	LONGTEXT	Boundary geometry

SQL

```
CREATE TABLE gadm_locations (  
  location_id INT AUTO_INCREMENT PRIMARY KEY,  
  country VARCHAR(100),  
  state_code VARCHAR(10),  
  state_name VARCHAR(100),  
  municipality VARCHAR(150),  
  geometry LONGTEXT  
);
```

Table 2: dengue_cases

Stores dengue case records.

Column	Data Type
dengue_id	INT AUTO_INCREMENT PRIMARY KEY
location_id	INT
report_date	DATE
year	INT
month	INT
cases	INT

SQL

```
CREATE TABLE dengue_cases (
  dengue_id INT AUTO_INCREMENT PRIMARY KEY,
  location_id INT,
  report_date DATE,
  year INT,
  month INT,
  cases INT,
  FOREIGN KEY (location_id)
  REFERENCES gadm_locations(location_id)
);
```

Table 3: malaria_cases

Stores malaria case records.

Column	Data Type
malaria_id	INT AUTO_INCREMENT PRIMARY KEY
location_id	INT
notification_date	DATE
year	INT
month	INT
laboratory_result	VARCHAR(50)

SQL

```
CREATE TABLE malaria_cases (
  malaria_id INT AUTO_INCREMENT PRIMARY KEY,
  location_id INT,
  notification_date DATE,
  year INT,
  month INT,
  laboratory_result VARCHAR(50),
  FOREIGN KEY (location_id)
  REFERENCES gadm_locations(location_id)
);
```

Table 4: precipitation

Stores rainfall information from the CHIRPS dataset.

Column	Data Type
precipitation_id	INT AUTO_INCREMENT PRIMARY KEY
location_id	INT
observation_date	DATE
year	INT
month	INT
rainfall_mm	DECIMAL(8,2)

SQL

```
CREATE TABLE precipitation (  
  precipitation_id INT AUTO_INCREMENT PRIMARY KEY,  
  location_id INT,  
  observation_date DATE,  
  year INT,  
  month INT,  
  rainfall_mm DECIMAL(8,2),  
  FOREIGN KEY (location_id)  
  REFERENCES gadm_locations(location_id)  
);
```

Table 5: temperature

Stores land surface temperature information from the CHIRTS dataset.

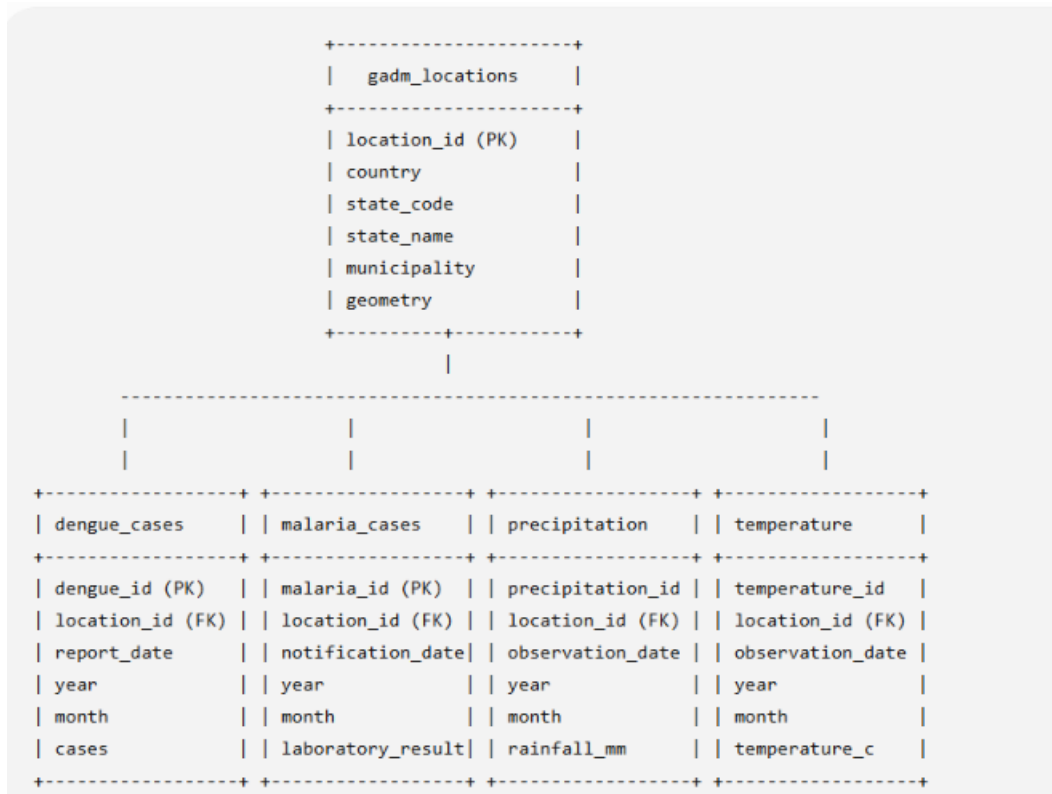
Column	Data Type
temperature_id	INT AUTO_INCREMENT PRIMARY KEY
location_id	INT
observation_date	DATE
year	INT
month	INT
temperature_c	DECIMAL(5,2)

SQL

```
CREATE TABLE temperature (  
  temperature_id INT AUTO_INCREMENT PRIMARY KEY,  
  location_id INT,  
  observation_date DATE,  
  year INT,  
  month INT,  
  temperature_c DECIMAL(5,2),  
  FOREIGN KEY (location_id)
```

```
REFERENCES gadm_locations(location_id)
);
```

Entity Relationship Diagram (ERD)



Relationships

Parent Table Child Table Relationship

gadm_locations dengue_cases One-to-Many

gadm_locations malaria_cases One-to-Many

gadm_locations precipitation One-to-Many

gadm_locations temperature One-to-Many

A single municipality (location) can have many dengue records, many malaria records, many precipitation records, and many temperature observations over time.

Why this design

- **No duplicated location data.** Geographic information is stored only once in the `gadm_locations` table.
- **Easy to join datasets.** Disease, rainfall, and temperature can be compared by municipality and date.
- **Normalized structure.** State and municipality names are not repeated in every table.
- **Scalable.** Future datasets (population, humidity, vegetation index, air quality, etc.) can be added as new tables linked through `location_id`.
- **Dashboard-ready.** This structure is efficient for creating visualizations, trends, and comparisons for your Brazil disease analysis dashboard.